STACKS, DERIVED CATEGORIES OF SPACES: UNBOUNDED-MAYER-VIETORIS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2067-2086 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 2088-2116, the complete author proof (27 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 12 of queue_spaceperfect.json (pb=1078, stmt 2067-2086, proof 2088-2116).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. gate_ok=true: raw-interval guard false positive (public-items.jsonl upstream-frame coords); precard_gate.py shift-corrected check cleared this label 2026-10-14 (gate_sp01.sh).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-unbounded-mayer-vietoris' also exists in stacks-cohomology — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0152: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2116–2116. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 014Z: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2116–2116. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 079P: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2116–2116. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08GW: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1889–1909. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $(U \subset X, V \to X)$ be an elementary distinguished square of algebraic spaces over S. For an object E of $D(\mathcal{O}_X)$ we have a distinguished triangle*

$$R\Gamma(X, E) \to R\Gamma(U, E) \oplus R\Gamma(V, E) \to R\Gamma(U \times_X V, E) \to R\Gamma(X, E)[1]$$

and in particular a long exact cohomology sequence

$$\ldots \to H^n(X, E) \to H^n(U, E) \oplus H^n(V, E) \to H^n(U \times_X V, E) \to H^{n+1}(X, E) \to \ldots$$

The construction of the distinguished triangle and the long exact sequence is functorial in E.

Proof. Choose a K-injective complex $\mathcal{I}^\bullet$ representing E whose terms $\mathcal{I}^n$ are injective objects of $Mod(\mathcal{O}_X)$, see Injectives, Theorem 079P. In the proof of Lemma 08GW we found a short exact sequence of complexes

$$0 \to \mathcal{I}^\bullet \to j_{U,*}\mathcal{I}^\bullet|_U \oplus j_{V,*}\mathcal{I}^\bullet|_V \to j_{U\times_X V,*}\mathcal{I}^\bullet|_{U\times_X V} \to 0$$

Since $H^1(X, \mathcal{I}^n) = 0$, we see that taking global sections gives an exact sequence of complexes

$$0 \to \Gamma(X, \mathcal{I}^\bullet) \to \Gamma(U, \mathcal{I}^\bullet) \oplus \Gamma(V, \mathcal{I}^\bullet) \to \Gamma(U \times_X V, \mathcal{I}^\bullet) \to 0$$

Since these complexes represent $R\Gamma(X, E)$, $R\Gamma(U, E)$, $R\Gamma(V, E)$, and $R\Gamma(U \times_X V, E)$ we get a distinguished triangle by Derived Categories, Section 014Z and especially Lemma 0152. □